

Study Selection

Records identified through the searches were combined and deduplicated before screening. Two reviewers independently screened titles and abstracts according to the prespecified eligibility criteria. Records with uncertain eligibility were retained for full-text assessment rather than excluded during initial screening.

Potentially eligible reports then underwent independent full-text assessment by two reviewers. One primary reason for exclusion was recorded for each report excluded at the full-text stage. Disagreements were resolved through discussion and re-review of the source report. Decisions that remained unresolved were adjudicated by the third author. Multiple reports arising from the same underlying study were identified and linked to avoid double-counting.

The study-selection process was documented using a PRISMA 2020 flow diagram. The searches identified 46,916 records from databases and other methods. After the removal of 21,692 duplicate records and 12 other records before screening, 25,212 records underwent title and abstract screening. Of these, 24,884 were excluded, and 328 reports were sought for retrieval. Seven reports could not be retrieved, leaving 321 reports for full-text eligibility assessment. Following the exclusion of 283 reports, 38 studies met the eligibility criteria and were included in the systematic review. Reasons for exclusion at the full-text stage are presented in the PRISMA 2020 flow diagram (Figure 1).

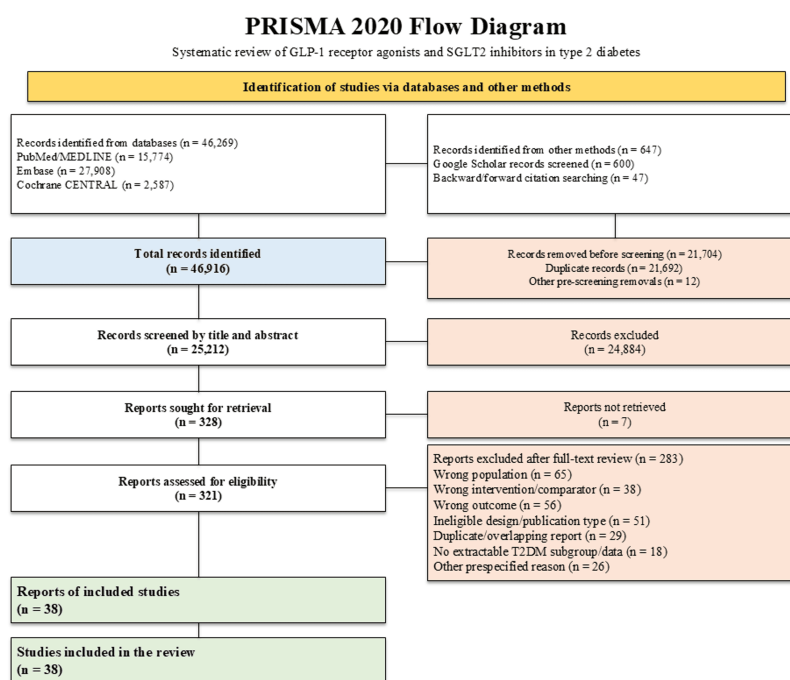


FIGURE 1: PRISMA 2020 flow diagram of study identification, screening, eligibility, and inclusion.

PRISMA 2020 flow diagram depicting the identification, screening, full-text assessment, and inclusion of studies in the systematic review evaluating GLP-1 RAs and SGLT2 inhibitors in T2DM. Records were identified through PubMed/MEDLINE, Embase, Cochrane CENTRAL, Google Scholar, and backward and forward citation searching. After duplicate and other pre-screening removals, 25,212 records underwent title and abstract screening, 321 reports were assessed for eligibility, and 38 studies were included in the final review. Reasons for full-text exclusion are shown in the diagram. The figure was created using Microsoft PowerPoint (Microsoft Corporation, Redmond, WA).

GLP-1 RAs: Glucagon-like peptide-1 receptor agonists; PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses; SGLT2: Sodium-glucose cotransporter 2; T2DM: Type 2 diabetes mellitus.

Data Extraction

Data were extracted using a structured approach. Extracted variables included study citation, study design, setting, sample size, participant characteristics, intervention and comparator, dose when relevant, duration of follow-up, baseline ASCVD, heart failure and CKD status, reported outcomes, effect estimates and 95% confidence intervals when available, adverse events, funding information, and information required for risk-of-bias assessment.